

Quarto Code Annotations Guide

Complete Reference for Teaching Scripts

Introduction

This document explains the Quarto features and R code elements used in graduate-level data science teaching scripts. It is designed as a comprehensive instructional reference for students learning reproducible research workflows.

1. Basic R Code Chunk Structure

```
```{r}
#| label: chunk-name
#| message: false
#| warning: false
R code goes here
```
```

Chunk options control how code behaves during rendering. Labels allow cross-referencing, and execution controls determine whether code is shown or run.

2. Common Chunk Options

- `label`: Unique identifier required for cross-references (fig-, tbl-, eq-).
- `message: false` — suppresses package startup messages.
- `warning: false` — suppresses warning output.
- `echo: false` — hides code in the rendered document.
- `eval: false` — shows code without executing it.
- `fig-cap` — adds a caption below figures.
- `tbl-cap` — adds a caption above tables.
- `fig-width / fig-height` — control figure size (in inches).
- `layout-ncol` — arrange multiple figures in columns.

3. Inline R Code

```
The mean richness is `r mean(data$richness)`.
```

Inline R evaluates expressions directly inside text. This ensures reported values update automatically when data change.

4. Cross-References

- Label figures: #| label: fig-name
- Reference in text: @fig-name
- Label tables: #| label: tbl-name
- Label equations: {#eq-name}
- Do not manually type figure or table numbers.

5. Callout Blocks

```
:::{.callout-note}  
## Sampling Period  
Data collected June–August 2024.  
:::
```

Callouts highlight important contextual information. Types include note, warning, tip, and important.

6. Layout Features

```
:::: {.columns}  
  
::: {.column width="50%"}  
Left column  
:::  
  
::: {.column width="50%"}  
Right column  
:::  
  
::::
```

7. Common R Functions Used

- `mean()`, `sd()`, `round()` — descriptive statistics.
- `aov()`, `summary()` — statistical modeling.
- `group_by()`, `summarise()`, `filter()`, `mutate()` — tidyverse data manipulation.
- `ggplot()`, `geom_point()`, `geom_boxplot()`, `geom_smooth()` — visualization.
- `kable()`, `kable_styling()` — formatted tables.
- `sessionInfo()` — record reproducibility information.

8. Best Practices Checklist

- Use descriptive, consistent chunk labels.
- Write complete-sentence captions.
- Use cross-references rather than manual numbering.
- Ensure the document renders without errors before submission.
- Maintain reproducibility by including session information.